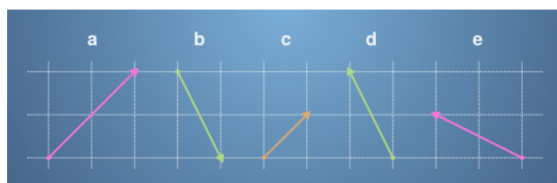


1. This aim of this quiz is to familiarise yourself with vectors and some basic vector operations.

1 point

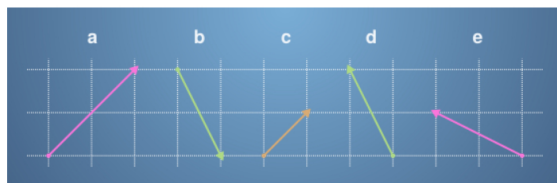
For the following questions, the vectors **a**, **b**, **c**, **d** and **e** refer to those in this diagram:



The sides of each square on the grid are of length 1. What is the numerical representation of the vector **a**?

- ☐ $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- ☒ $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

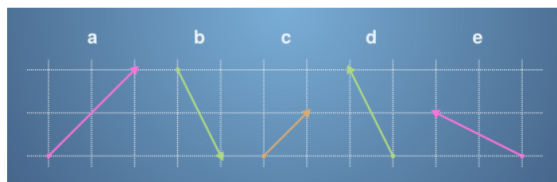
1 point



2. Which vector in the diagram corresponds to $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$?

- ☐ Vector **a**
- ☐ Vector **b**
- ☐ Vector **c**
- ☒ Vector **d**

1 point



3. What vector is $2\mathbf{c}$?

Please select all correct answers.

- ☐ **e**
- ☒ **a**
- ☐ $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$
- ☒ $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$

1 point



4. What vector is $-\mathbf{b}$?

Please select all correct answers.

☒ $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

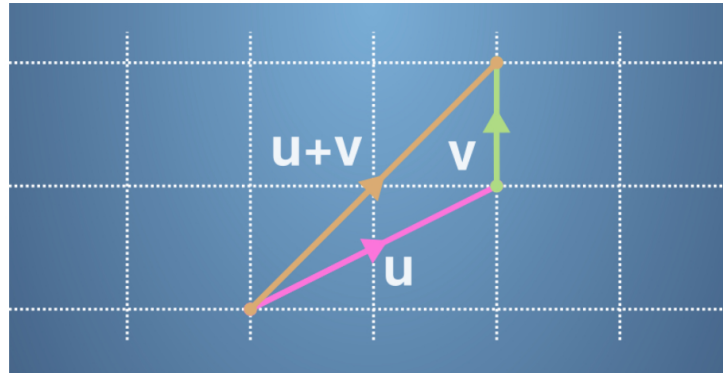
☒ $\mathbf{d}$

☐ $\mathbf{e}$

☐ $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

5. In the previous videos you saw that vectors can be added by placing them start-to-end. For example, the following diagram represents the sum of two new vectors, $\mathbf{u} + \mathbf{v}$:

1 point



The sides of each square on the grid are still of length 1. Which of the following equations does the diagram represent?

☐ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

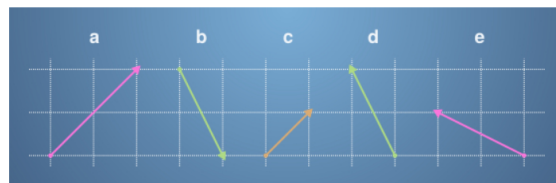
☐ $\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

☒ $\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

6. Let's return to our vectors defined by the diagram below:

1 point



What is the vector $\mathbf{b} + \mathbf{e}$?

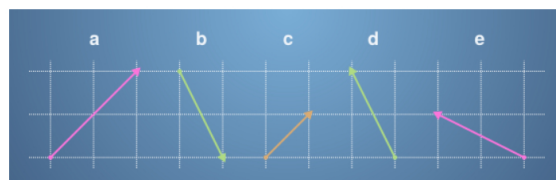
☐ $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

☒ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

☐ $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

1 point



7. What is the vector $\mathbf{d} - \mathbf{b}$?

1 point

- ☐ $\begin{bmatrix} 4 \\ -2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 2 \\ -4 \end{bmatrix}$
- ☐ $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$
- ☒ $\begin{bmatrix} -2 \\ 4 \end{bmatrix}$